Fig. 3 | Performance of RNA velocity methods on conventional scRNA-seq datasets. a, High-dimensional cell-level velocity cosine across 42 simulated datasets, shown with dataset-type and seed strips, median/mean summaries and a per-dataset mean row. **b,** Mean high- and low-dimensional cosine by method, with standard deviations. **c,** Pooled high- and low-dimensional distributions: 740 finite method–dataset pairs across 18 configurations, including deterministic Dynamo. Panel b and Extended Data Fig. 5a,b exclude this configuration and use 698 finite pairs across 17 configurations. **d,** Cosine agreement between gene-space and PCA performance-score vectors. Topology curves average the three per-dataset score-vector cosines; the total curve pools all 42 datasets before calculating that cosine. Extended Data Fig. 5c separately reports median projected-velocity accuracy against simulated truth. Both analyses exclude deterministic Dynamo. **e,** Schematic of the evaluation metrics. **f,** Pearson correlations among metrics. **g,** GDC distributions across 176 HuBMAP datasets, ordered by median; Best performance is the best available score across methods for each dataset. Boxes show quartiles, center lines show medians, and whiskers extend to 1.5 times the interquartile range. **h,** GDC across ten classic datasets, with the best available value for each dataset in the final column.